

Fariyaa

Machine Learning Engineer | Generative AI, LLMOps & MLOps

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PROFESSIONAL SUMMARY

Senior Data Scientist and Machine Learning Engineer with experience building scalable AI/ML systems, Generative AI applications, LLM-powered products, and production-ready MLOps and LLMOps pipelines. Skilled in designing automated ML workflows for data preprocessing, feature engineering, model training, evaluation, deployment, and monitoring across AWS, GCP, and Azure. Proven ability to collaborate with cross-functional teams, translate business challenges into statistical and deep learning solutions, and deliver enterprise-grade AI systems including RAG pipelines, conversational AI, recommendation engines, predictive analytics, and multi-agent architectures.

CORE SKILLS

Machine Learning • Deep Learning • Generative AI • Large Language Models • Natural Language Processing • Prompt Engineering • Fine-Tuning • Retrieval-Augmented Generation • Agentic AI • Autonomous AI Systems • Feature Engineering • Data Augmentation • Model Deployment • Model Evaluation • Hyperparameter Tuning • MLOps • LLMOps • AutoML • Model Serving • Data Science • Predictive Modeling • Statistical Analysis • Time Series Analysis • Recommendation Systems • Computer Vision • Data Pipelines • Data Processing • Analytics Reporting • Business Intelligence • Python • Scikit-Learn • TensorFlow • Keras • PyTorch • Hugging Face Transformers • OpenCV • NLTK • Pandas • NumPy • PySpark • FastAPI • Flask • Streamlit • LangChain • LangGraph • LlamaIndex • AutoGen • LangSmith • MLflow • ZenML • Docker • GitHub Actions • CI/CD • REST APIs • GraphQL • SQL • NoSQL • Vector Databases • Pinecone • Faiss • Chroma DB • Neo4j • AWS SageMaker • AWS Bedrock • AWS Lambda • AWS EC2 • GCP Vertex AI • BigQuery • Cloud Run • Azure ML • Azure AI • Azure App Services • Agile Methodologies • Stakeholder Management • Cross-Functional Collaboration • Team Leadership

PROFESSIONAL EXPERIENCE

Senior Data Scientist & Machine Learning Software Engineer, Generative AI | PURELOGICS

12/2024 - Present

- Spearheaded enterprise AI MVPs including RAG pipelines for BI, remote patient monitoring AI, GPT-4o conversational chatbots, and agentic AI systems for legal and financial advisory use cases.
- Designed, deployed, and optimized end-to-end AI/ML pipelines on AWS SageMaker, Bedrock, Lambda, GCP Vertex AI, BigQuery, and Cloud Run to support scalable production AI solutions.
- Engineered multi-agent architectures using LangChain, LangGraph, PhiData, Pinecone, and graph databases to improve automation, decision-making, and autonomous workflow execution.
- Led cross-functional delivery of MLOps and LLMOps pipelines, supporting model lifecycle management from experimentation through production deployment.
- Mentored AI/ML teams on LLM fine-tuning, Retrieval-Augmented Generation, prompt engineering, and AI-powered process automation.
- Delivered enterprise-grade AI solutions that improved deployment speed, business intelligence capabilities, and workflow efficiency.

Senior Data Scientist, Machine Learning & Gen AI Engineer | NVIDIA

08/2024 - Present

- Contributed to NVIDIA and Z by HP Developer Programs focused on GPU-optimized applications and enterprise Generative AI solutions.
- Developed and tested open-source and enterprise LLM solutions, including vLLMs, for Generative AI use cases.
- Integrated NVIDIA tools and SDKs to build, validate, and optimize AI solutions for large-scale deployments.
- Collaborated with NVIDIA and Z by HP teams to evaluate AI tools, frameworks, and deployment workflows.
- Supported open-source LLM initiatives and community innovation in Generative AI.

Senior Data Scientist & Machine Learning Engineer | BLOCBELT

12/2022 - 12/2024

- Led development of AI-driven SaaS and PaaS products, managing a team of three across big data, analytics, and AI engineering initiatives.
- Delivered AI solutions that generated approximately \$2M in client cost savings through automation, data processing optimization, and workflow improvement.
- Improved sales forecasting accuracy by 80-90% through predictive modeling and data analysis across 120+ market entities.
- Built and launched a personalized fashion e-commerce platform with a recommendation engine, increasing sales by 80% and online orders by 200%.
- Developed statistical models and predictive analytics that contributed to an 18% revenue increase and improved business decision-making.
- Enhanced data visibility by 45% through dashboards and analytics reporting for cross-functional stakeholders.
- Conducted 25+ A/B tests to optimize digital commerce experiences and improve SaaS and PaaS product performance.

AI Data Scientist Kaggle Master Mentor | GOOGLE-KAGGLE

12/2022 - 11/2024

- Mentored aspiring data scientists through the Kaggle-X BIPOC Mentorship Program, dedicating 20 hours per week to technical guidance and project development.
- Led a team of four mentees in machine learning, data science workflows, and practical problem-solving.
- Collaborated on high-impact AI projects and contributed to the global data science community.
- Recognized among the Top 100 Kaggle contributors globally.

Data Scientist | Crypto-Express

01/2022 - 12/2022

- Developed an anti-spoofing face application to improve digital identity security and reduce identity fraud by 70%.
- Applied machine learning and deep learning techniques to industry-critical problems, improving efficiency by 25% and reducing operational costs by 15%.
- Engineered predictive models and actionable insights that supported a 20% revenue increase.
- Performed exploratory data analysis, model development, and automation that saved 250+ hours of manual analysis annually.
- Delivered AI/ML projects in a remote environment while adapting to changing business and technical requirements.

Artificial Intelligence Engineer | Pakistan Freelancing Training Center

01/2021 - 12/2021

- Led AI and data science training sessions focused on practical machine learning applications and real-world problem-solving.
- Designed and deployed models using TensorFlow and Keras to improve system efficiency and prediction accuracy.

- Automated repetitive data tasks with Python, reducing manual effort and error rates.
- Conducted applied AI research to incorporate emerging methodologies into training and project work.

PROJECTS

AutoML-Studio

Built a low-code/no-code machine learning platform for rapid experimentation and deployment of classification, regression, time-series forecasting, and clustering models. Integrated MLflow for experiment tracking, SHAP for explainability, drift detection for reliability, and FastAPI and Streamlit for deployment.

Conversational AI Chatbot for Customer Support

Designed and deployed an enterprise-grade GPT-4o chatbot using LangChain and AWS Bedrock. Implemented intent recognition and dynamic response generation, reducing customer query resolution time by 40%.

Health Assistance Application

Architected an LLM-powered health assistance application using Gemini 1.5 Pro, RAG, and LangChain to provide medical information, symptom guidance, treatment suggestions, and preventive healthcare recommendations.

AI-Powered Blog Generator

Developed a content generation system using Llama 3.1 8B, AWS Bedrock, RAG, and Streamlit to automate blog creation, editing, and publishing with real-time adaptation to dynamic datasets.

DocuWiz AI

Engineered a document intelligence tool using BART, DistilBERT, RAG, and LLMs for extraction, summarization, sentiment analysis, and visualization across PDF and DOCX files. Reduced document processing time by 60%.

Customer Satisfaction Prediction System

Created a predictive analytics system using ZenML and MLflow within an MLOps framework, achieving 93% accuracy in forecasting customer satisfaction and supporting improved retention strategies.

Harvestify AI-Powered Plant Health Assistant

Designed an Azure-based computer vision application for real-time plant leaf disease diagnosis, crop recommendations, and fertilizer suggestions. Reduced disease detection time by 70% and improved crop yield predictions by 30%.

EDUCATION

Bachelor in Computer Science, Major: Artificial Intelligence and Computer Science | Institute of Management Sciences

CERTIFICATIONS

- IBM Machine Learning Specialization Professional Certificate, IBM
- Deep Learning Specialization, DeepLearning.ai
- Machine Learning Engineering for Production (MLOps), DeepLearning.ai
- MLOps | Machine Learning Operations Specialization, Duke University
- Large Language Model Operations (LLMOps), Duke University
- Practical Data Science on the AWS Cloud Specialization, DeepLearning.ai
- Advanced Machine Learning on Google Cloud Specialization, Google Cloud
- Preparing for Google Cloud Certification: Machine Learning Engineer, Google
- Microsoft Azure AI Fundamentals AI-900 Exam Prep Specialization, Microsoft
- AWS Cloud Solutions Architect Professional Certificate, Amazon Web Services

- Prompt Engineering for ChatGPT, Vanderbilt University
- Machine Learning Specialization, DeepLearning.AI
- Generative Adversarial Networks (GANs) Specialization, DeepLearning.ai
- Google Advanced Data Analytics Professional Certificate, Google
- Google Business Intelligence Professional Certificate, Google
- AI Product Management, Duke University

ACHIEVEMENTS

- Top 100 Global Kaggle Master contributor
- KaggleX BIPOC Mentor
- 24x LinkedIn Top Voice
- Contributor to NVIDIA Developer Program
- Contributor to Google Developer Group
- Contributor to AWS AI Community
- Recognized as the first Pakistani mentor in the Kaggle-X BIPOC Mentorship Program